Family Names as Traces of the Past: Exploring the Potentials of the Census of 1956 for the Study of Acculturation in Taiwan

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Abstract

When the ROC state in 1956 conducted an ethnic census, with the obvious political purpose to show that, as Minister HSU puts it, "The Taiwanese are part of the Chinese nation ..." and "all of the surnames of Taiwanese originated in the middle area of mainland China" , the Minister couldn't know that the resource created from this census actually would prove the opposite. In fact, in many regions of Taiwan, there is a striking similarity of Indigenous surname distributions (assigned by Qing and Japanese officials) and Holo and Hakka surname distribution, which might be explained by a process of acculturation in general and sinicization more particularly. Starting from these observations, we explore the census of 1956 in its digitized form, to see in how far it can inform us about acculturation processes that took place in late Japanese period and early ROC period. Using resampling methods as statistical technique, we can simulate migration and acculturation processes and check whether locally observed surname distributions may have been achieved by these processes. To validate the outcomes of these simulations, we explore the mortuary practices of those families that regionally are identified as having undergone an acculturation, to see whether features of culture of origin can still be found in our times.

Introducton

The ThakBong Project

The ThakBong Digital Archive Project[1], aims at a representative digital documentation of Taiwan's gravesites[2]. Gravesites are modeled as structure composed of tombs, with as central element the tombstone, graveyard markers, landscapes, temples, bone-ash-towers, crematoriums and links to local communities. The documentation is mainly based on a large number of geo-referenced images of these objects, to which we add digital maps, satellite images, video segments, interview recordings or short background stories.

From the primary image and GPS data, we derive maps and elevation models of the gravesites, transcriptions, transliterations, syntactic and semantic analyses of the transcribed inscriptions, featurized classifications of photographed objects and listings of offerings, symbols attached to the tomb and components of figurative representations.

The objects in their digitized form become accessible through key-words like family names, the life data of a deceased, placenames written on tombstones, symbols, offerings or the architecture of the tomb. Through the search of two or more features, correlations between features can be established, which represents a basic methodological approach in the analysis of such digital data. The correlation of such features with spatial data allows for geographic analyses of cultural variations, representable in maps. A time-line analyses of tombs and gravesites can be obtained through the correlation of features with the temporal indicators on tombstones, showing how tombs change in time in reaction to various events.

Up to now, about 40.000 tombs of more than 500 gravesites have been documented through 140.000 photos. The fieldwork, the data collection, the processing of the data and the annotation are permanently ongoing in parallel. Due to the rapid transformation of Taiwan's society, its landscapes and cultural practices, tombs and graveyards are regularly removed under the pressure of private investors and political conformism. The documentation word can thus be conducted only in a narrow time-window, spanning from now only a few years into the future. The sampling of the gravesites tries to be representative for Taiwan. As a standard for comparison we use the the ethnic and

1 From Taiwanese 讀墓 thak-bong ("study tombs")

2 We prefer the term *gravesite*, meaning a place where one or many tombs are located, over the term *graveyard*, as this latter term would evoke an organized ensemble of tombs, which in the case of Taiwan, is frequently not the case and has no tradition. Only gravesites set up recently by local communities might be comparable to organized western-style graveyards.

regional proportions of Taiwan's census of 1956, as represented in Chen and Fried (1968).

Outside Taiwan, the ThakBong archive documents gravesites in the People's Republic of China (PRC) and places where Chinese settlers have moved to, e.g. Hong Kong, Hawaii, USA, the Philippines and Europe. In China, our documentation effort centers on Wenzhou, where we have taken about 6.000 tombs of 2.500 tombs. As the size of these data set still don't compare to the data set on Taiwan, one has to be cautious about what interpretations one can draw from them. In most cases, it is epistemologically not sound to make positive or negative statement about these regions, for example, what is typical for China, what variation one might find there, or what cannot be found there.

The reason for this methodological cautiousness are the lessons the authors have learned from the work on the relatively small island of Taiwan. Even on this island, one finds a great local variation of cultural practices, and what is believed to be impossible and what nobody ever has heard about, can be found just in the next village on the other side of the road. How much greater must be the variation in larger spaces, such as China or in South-East Asia.

The smaller data set are not useless though. The comparison of the larger Taiwan set with the smaller sets yields shows similarities and dissimilarities. If for example, in the sample from China one finds recurrently a feature, which one never or rarely finds in the large Taiwan sample, this might indicate a difference. If, however, one finds recurrently a feature in the Taiwan set, but not or only rarely in the smaller China set, this might just be consequence of a limited size of the tomb sample from China. Epistemologically, a statement about the difference between two sets is weaker than a statement about the properties of each set alone. But frequently it might just be this difference that one wants to know about.

An image and its associated data, e.g. the annotations, transcriptions and descriptions, are made available under the AttributionNonCommercial-NoDerivs 2.5 Generic licence of CreativeCommons[3], as soon as the image has been annotated with a minimal set of features. The current web-site of the project, http://thakbong.dyndns.tv, allows for a visual appreciation of the data and the download of individual data and images for research purposes. For research that require the access to larger data-sets, we prepare the distribution of the entire data set under the form of easy-to-convert tables, which can be read in, among others, by spreadsheet programs. A written documentation of how to start research with these data-sets in R[4] is under preparation.
In addition to our proper channels of distribution, the *TELDAP* agency [5] distributes 10.000 ThakBong photos with their Dublin Core meta-data. These data have been created by the ThakBong team between 2010.08.01 and 2011.07.31, in an NSC-TELDAP project under grant number 99-2631-H-390-002-.

The 1956 Census

Visionary research allows for applications that few people had in mind at the time the research was conducted. The work of Chen and Fried (1968 and 1970) certainly belongs to this category of important and far-reaching work. When Chen and Fried worked in 1964, equipped with little more than paper and pencil, through millions of ten-year-old census sheets to produce a book filled on 969 pages with nothing but numbers, who but the researchers might not have thought of a waste of time and money? Attempts to exploit these data, in Pan (1987) and Pan (2008), struggled mainly with the size and the format of these paper-bound data.

Fifty years after the original compilation of the data, the authors of this paper have digitized the book into a computational data-set, see Streiter, Goudin, Huang and Lin (2011). Using a relational database as a computational backbone and an intelligent interface, which guided the input procedure, filled out implied numbers and performed all kinds of input control, all data could be captured within a month.

Linking the administrative classification used in 1956 within the census to our geo-spatial data on Taiwan, we obtained a powerful Geographic Information System, which successfully can be applied to many research questions. To distinguish the original data set from the digitized set, we call the latter, for our convenience, *e56*.

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3 http://creativecommons.org/licenses/by-nc-nd/2.5/

4 R is a free software environment for statistical computing and the creation of graphics, including maps. R is available for a wide variety of platforms and is much more powerful than spreadsheets are, see http://www.r-project.org/

5 http://teldap.tw/

臺灣省縣市鄉鎮別、姓氏

Distribution of Surnames by Town

I. 桃園縣 T'aoyüan Hsien

Characters 中文	Surnames 羅馬字 Romanized Form	I 桃園縣 T'aoyüan Hsien 合計 Total	本省 Taiwanese 福建 Fukien Origin	本省 Taiwanese 廣東 Kwang-tung Origin	本省 Taiwanese 其他 Others	山胞 Aborigines	外省籍 Main-landers	I-1 桃園鎮 合計 Total	I-1 桃園鎮 本省 福建 Fukien Origin
計	**Total**	**98,062**	**46,997**	**42,455**	**36**	**1,534**	**7,040**	**11,780**	**9,076**
陳	Ch'en²	8,237	4,671	3,020	1	136	409	1,177	998
林	Lin²	4,192	2,740	1,159	—	166	127	651	572
黃	Huang²	7,549	3,491	3,696	—	158	204	776	665
張	Chang¹	4,009	1,521	2,011	2	52	423	470	328
李	Li³	5,297	3,320	1,328	1	118	530	690	533
吳	Wu²	2,978	1,418	1,358	1	49	152	574	492
王	Wang²	2,463	1,256	755	—	67	385	317	179
劉	Liu²	3,316	1,055	1,906	2	11	342	428	214
蔡	Ts'ai⁴	1,009	791	170	—	2	46	158	142
楊	Yang²	2,683	1,814	573	—	65	231	640	595
許	Hsü³	2,518	1,817	649	1	15	36	219	201
鄭	Cheng⁴	1,334	822	442	—	4	66	260	232
謝	Hsieh⁴	1,874	520	1,268	—	7	79	125	83
郭	Kuo¹	949	824	46	1	4	74	133	117
洪	Hung²	254	174	29	—	31	20	51	43

1. Figure: Part of a page of Chen and Fried (1968) showing the surnames on the left, the administrative class on the top, the ethnic classification below and the size of the sample in the middle.

The interest in the geographic distribution of surnames in Taiwan, today, as in 1956, relates on the one hand to the emic use of surnames as an idealized compass in search of an identity. With the help of one character, a surname suggests to its bearer thousands of years of family history. The surname 粘 (Nié) for example, is generally associated with a migration from Manchuria through Fujian to Taiwan. Objectively, such a genealogic line excludes $(n\text{-}1)^2\text{-}1$ ancestors of n ancestral generations and thus provides space for narratives that tend to ignore non-canonical forms of

surname transmissions, inter-ethnic marriages, uxorilocal marriages or adoption within and across ethnic boundaries, see Brown (2004). As collective trait of communities, on the other hand, surnames actually preserve history, if not thousands of years, but longer than local practices related to languages, music, dietary habits and religions might do. A surname which is relatively frequent in a community will not easily disappear, no matter whether surnames are inherited along paternal or maternal lines, adopted children preserve their original surnames or take over the surname of the adoptive parents. Thus, a community which has 陳 (Chén) as its most common surname, has another history than a community where the most common surname are 邱 (Qiū), 鍾 (Zhōng) or 潘 (Pān). Surnames are thus fingerprints of communities, comparable also to genetic markers which tell stories, not of the individual but of a groups of people. Driven by the interest in surnames as collective traits, Chen and Fried sampled in 1964 one quarter of the 9.311.312 personal census cards of 1956. The cards had been stored after 1956 in 361 packages, lumping village and city communities together into townships and city districts. Data on Kinmen and Mazu are not reported.

The original data set and *e56* can be conceptualized as three related three-dimensional tables, which specify the same information at three levels: First Taiwan and Penghu, second counties and cities and third townships and city districts. Each table has three dimensions. The administrative classification, implied by the awkward physical packaging, formed the first dimension. 1027 previously attested Chinese surnames formed the second, and an ethnic classification the third dimension. The ethnic categories, which de-facto were frequently understood by officials as linguistic category, have been: *Fukien*, *Kwangtung*, *Mainlander*, *Indigenous*, *other*. Through the use of language as criterion of an ethnic classification, Holo-speaking Hakka and Holo-speaking Indigenous have to an unknown extent been classified as *Fukien*. The 1.855.540 table cells reports for each administrative unit, each ethno-linguistic classification and each of the 1027 surnames, the number of the sampled sheets, after removing three quarters of the original sheets. Multiplying these numbers by 4 gives an estimate, but not the exact number of the people classified in a certain way in 1956. This estimate might, for larger numbers, be close to the actual size of the so-classified population. Small numbers, especially numbers below ten, provide only a rough estimate for the classified population. As the individual persons are unknown, the interest of these tables lies in the characterization of local communities or groups of people sharing the same name. Generally speaking, given two variable values, one can obtain probabilities estimates for the third variable. For example, the variables *surname* and *location* instantiated as 陳 (Chén) and *Taipei* yields a probability distribution of ethnicity as: *Fukien*: 76%, *Kwangtung*: 2% and *Mainlander*: 21%.

The data reflect very much the time in which they have been gathered, the technical approach and the prevailing ideology. As the physical grouping of the census papers represented the administrative classification, and different villages have been stored together, this caused an unnecessarily broad administrative classification in the final data. From today's point of view, one can also barely understand why the researchers have removed three quarters of the data, as the manual removal, one by one, could not save much time compared to the recording of the ethnicity and the surname marked on the sheet on a piece of paper. Especially for low frequency surnames, the data loss is quite significant. Then, in 1964 no computer was used. Inconsistencies therefor showed up during the digitization in margins and higher level tables, which we corrected silently in *e56*, assuming the more elementary value to be the correct one and thus correcting the values in the margins. Finally, social tensions and strong ideologies in the time of the census left their imprint on the data.

First of all, the ethnic classification in the census foresees neither ethnic changes across generations nor mixed ethnic ancestry. The data are thus not objective but represent a strong emic Han view on how ethnicities classify. In Taiwan, however, Indigenous groups, Holo and Hakka have been constantly crossing ethnic boundaries through adoption, inter-ethnic marriage or acculturation. Second, as mentioned before, the language of the people was used as prevailing criterion for the ethnicity, and levelled out ethnic variations in a period, the KMT hoped to reconquer the Chinese mainland through the "Chinese nation"[6]. Holo-speaking Hakka and Pepo communities that had lost heir ancestral languages have, to different degrees and depending on the local administration, been classified as Fukien, as reported frequently by informants who have for comparison access to their older Japanese census data. How Holo-speaking Hakka have been classified is unclear. To understand how these more difficult classifications have been made, a more in-depth analysis of the data is required. Finally, the list of the 1027 surnames used in Taiwan is certainly a curtailed, if not ideologically cleaned list. It not only excludes extremely rare Mainlander surnames, see Yang

6 *"From this report of this study, we can further understand that Taiwanese are an inseparable part of the Chinese nation, and the work's contributions to the problem of census policy in Taiwan is so obvious that it is unnecessary for me to go into particulars"* Hsü (1968), Minster of Interior in the *Introduction* to Chen and Fried (1968).

(1979), but also non-standardized double surnames, see Streiter, Goudin, Huang and Lin (2011).

The ideological interest in the study of Chen and Fried (1968) was thus to provide a scientific foundation for the claims the ROC-state made on ethnicity and culture. For this reason the study received governmental support in its first edition and twenty years later, when it was reworked by Pan and reprinted by the Taiwan Province Government in 1987. These ideological purposes are made explicit in the introduction to the original publication by the Minister of the Interior. The ideological claims, in addition, also seem to be confirmed by a face-validation of the data: All people in Taiwan share the same stock of family names. This ideological conception of the time as condensed into one slogan in an almost parallel and influential publication in 2002 by Peng as *"Five Hundred Years Ago, It Was One Family"* .

How blind ideology can make, is exemplified once more through the census data of 1956. If politicians had really looked at the data, they might not have agreed to their publication. Even inspections with the naked eye reveal that long lists of low-frequency surnames are restricted to one category only, *Mainlander*, which in this sample represent a minority of 9.8%. More than the half of all *Mainlander* surnames, that is 447 surnames, are not shared by any other local Han group, which after all make up 87% of the census population. The minority of Mainlander covers 868 surnames, while the majority of Taiwanese Han groups covers only 553 surnames. Thus, while Taiwanese Han share their surnames with the Mainlander, Mainlander don't share their names with Taiwanese Han. They are foreigners. In addition, an unexpected similarity of the categories *Mainlander* and *Indigenous* is apparent. 60 of the 868 *Mainlander* surnames, are shared exclusively with the marginal group *Indigenous* of only 2.3%. The more hidden secrets of these data however cannot be revealed by the naked eye and require the intervention of computational devices. It is this the main contribution of this publication.

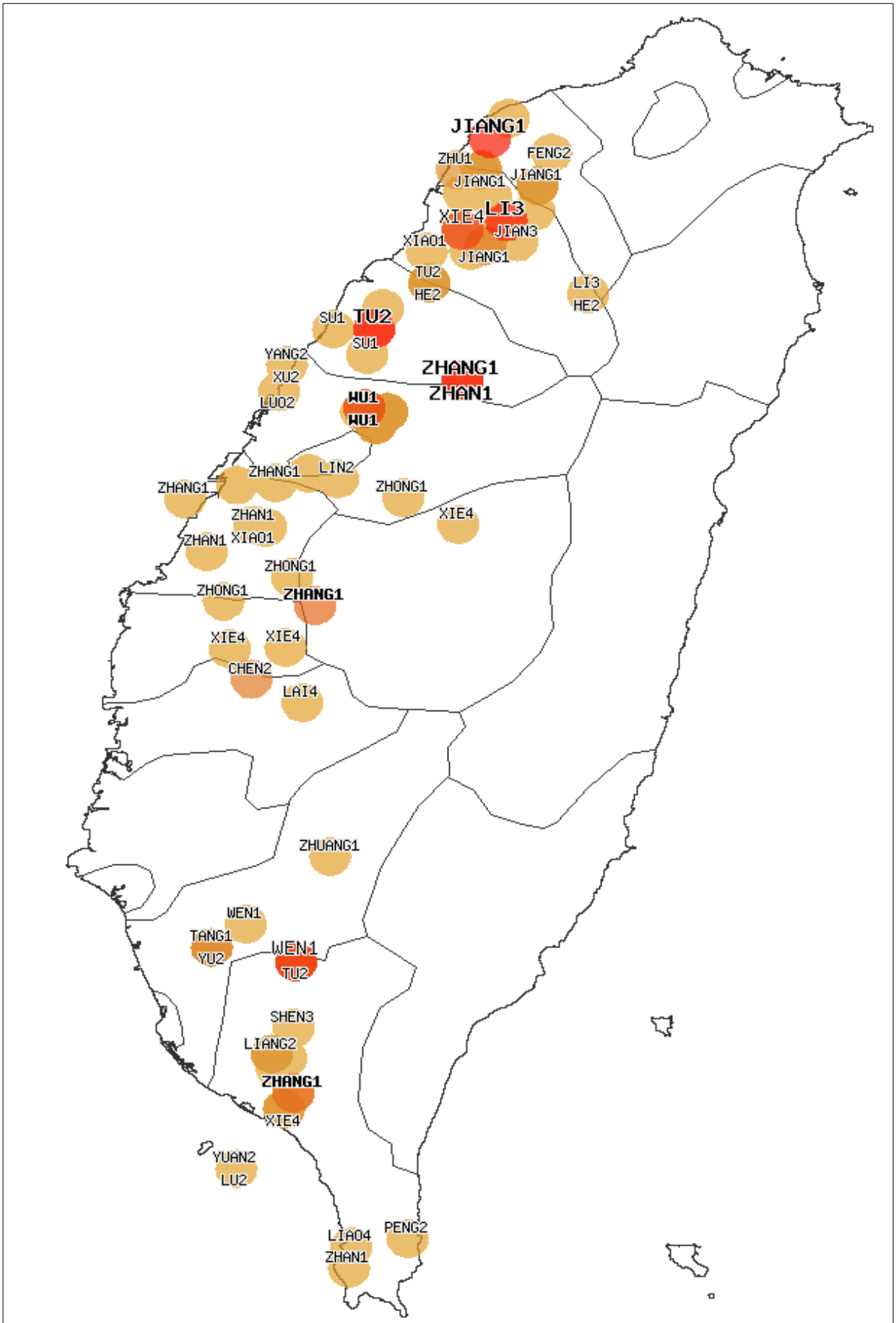
Given the three-dimensional nature of the tables and their huge size, these usage of these data has been prohibitively time consuming when still bound to paper. What is logically a tree-dimensional table, is restricted on paper to one of three logically possible views. Given the surname and the administrative unit, the ethnic distribution can be relatively easily found. To obtain, however, the administrative distribution for 陳 (Chén) and *Kwangtung*, one has to collect data from 90 pages of the book. For low-frequency surnames, the number of pages to be consulted may rise to 361. Thus, not one, but three volumes a 1000 pages would have been required to represent all possible views. Probably as a compensation for the difficult accessibility of the data, Chen and Fried produced a second Volume, which shows on maps the spatial distribution of some surnames under certain ethno-linguistic parameters. However, if the data used to produce maps are not given with the maps, the maps remain purely impressionistic and cannot be reused for further research. The first exploitation of the data we have found, twenty years after the original publication in Pan (1987). In this publication, sponsored by the Taiwan Province Government, thus not free from ideological intentions, Pan (1987) reproduced the main data body of the original resource, plus in addition some percentages on the margins. Ironically, what might have been a complex and tedious work at that time, can be done today with a computer in a few second.

Another twenty years later, the same author, Pan (2008), made in Pan (2008) another recompilation of the original data, comparing the composition of communities of 1956 with implicitly given composition of communities today. The main line of the author's argumentation is the following: If in a certain township in 1956, people with a certain surname where classified as *Indigenous*, and nowadays this township no longer has an Indigenous population, then people with that surname in that township are likely to be accultured Indigenous. If in addition, this township never had an indigenous population belonging to one of the ethnicities recognized by the ROC, such as Ami or Bunun, then the indigenous recorded in the census of 1956 were Pepo communities[7]. The data in 1956 thus can be read as specifying the probability that a person in a certain township with a certain surname had ancestors who still in 1956 were Pepo. The recompilation of the original data of Chen and Fried (1968) by Pan (2008) thus cut out of the total data set former Pepo townships and Pepo surnames.

In their computational form of *e56*, the size of the three dimensional tables is no longer an obstacle. To the contrary, the huge amount of data form the ingredients for a rich research resource. After the digitization of the census data, the authors have tried to tackle a question, related to that of Pan (2008), but in its methods and implications quite different. While Pan (2008) took the census data as a historic fact which compares to the present, we compare the census data to its past. The central and intriguing question is, who of the people marked as *Fukien* in this census, have been actually Pepo at the time of the census or some decades before. While a first attempt to tackle this question has been made in Streiter, Goudin, Huang and Lin (2011), this paper tries to bring about

7 meaning *"Flat Land Aborigines"*.

considerable progress in this question.



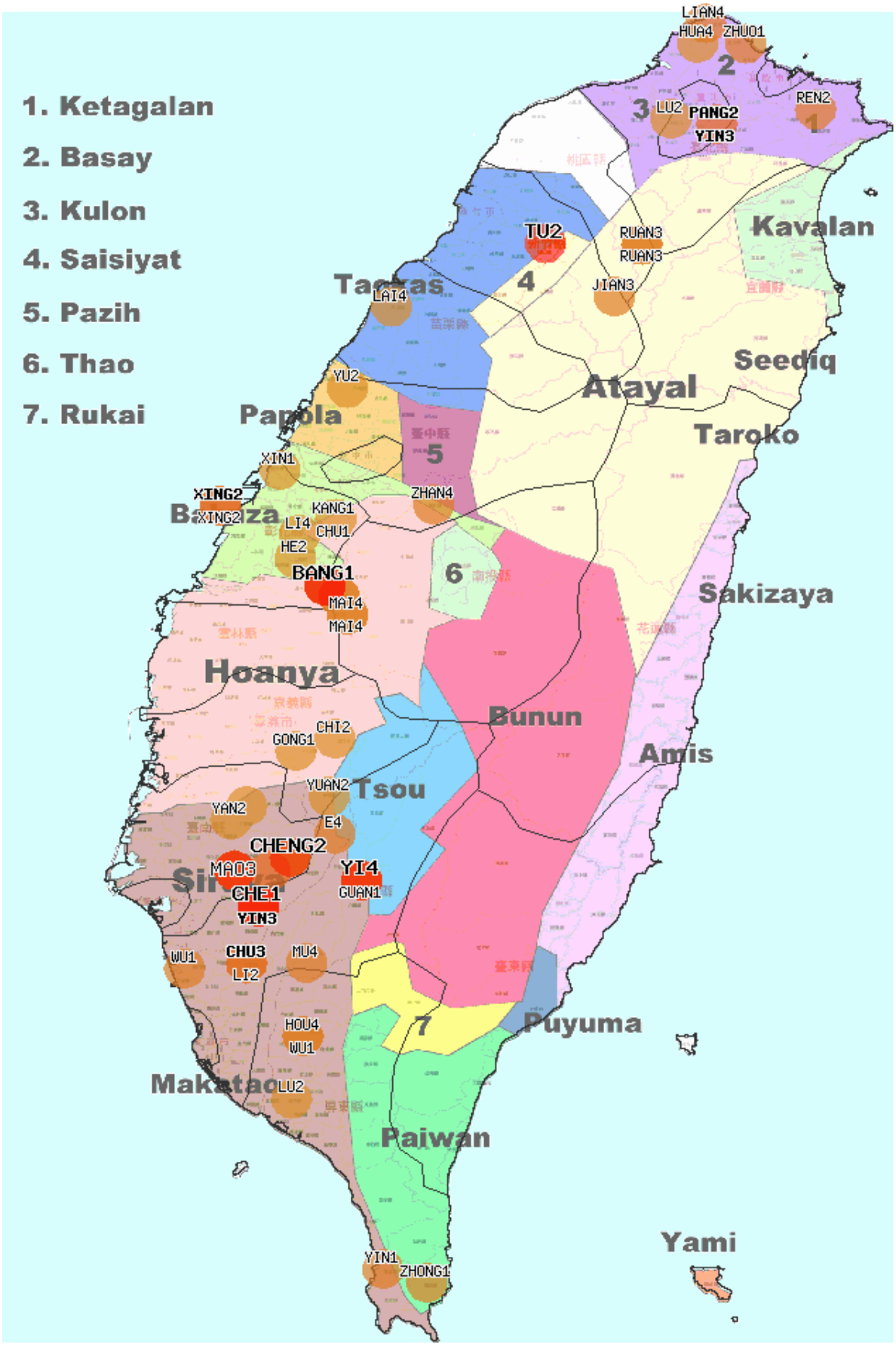
2. Figure: Locally unexpected many Holo surnames which are local common Hakka names.

The map in Figure reproduces the data on potential Hakka to Holo acculturation of Streiter, Goudin, Huang and Lin (2011).

After the creation of *e56*, the authors have tried to tackle the question of whether the data can reveal, who of the people marked as *Fukien* have been Pepo of Hakka at the time of or before the

census, see Streiter, Goudin, Huang and Lin (2011). To approach this question, we identified those Holo surnames that were frequent in a township but highly unexpected for a Holo community in general. As for the classification of Hakka that had been classified as Holo, we extracted from the unexpected list of Holo names, those surnames that were frequent among the Hakka of the same township, assuming that the high frequent Hakka surnames had become through the process of acculturation the source for the unexpected Holo surnames. The remaining unexpected Holo surnames were then assumed to be of Indigenous origin. Although we could produce maps which seemed to show reasonable results, this technique did not result in any quantification. Neither the significance of a reclassification of ethnic label of 1956 nor a probability model can be calculated from that. As the final filtering of the outcomes cannot rely on confindence intervals, arbitrary threshold values had to be used, which make the approach quite unreplicable. Nevertheless we could show that unexpected frequencies are one way to capture anormalities which are probably caused by an acculturation that preceded the census.

A question we thus pursue in this paper, is how questions of acculturation can be tackled in a framework that allows for numeric estimates. What seems to be clear however from the 2 models discussed so far, is that different ethnicities, different communities and different time periods may require different models. Thus, different models may be required to identify those surnames that in 1956 where classified as Holo, depending on whether one wants to relate them to the Japanese or the late Qing period.



3. Figure: Locally unexpectedly high-frequent Holo surnames in the 1956 census which do not correspond to high-frequent Hakka names and are thus potentially caused by an acculturation of Indigenous groups.

Models to Estimate Ethnicity in ThakBong

Model I

Many of the surnames created by Qing officials have disappeared in Taiwan. Of those, which are still in use today, however, census data list them frequently as Holo or Hakka names. In 1826 for example, the surname 高 (Gaō) was assigned, according to Ino (1997), to Indigenous groups in Danshui. In 1956, according to the census of that year, there were in Danshui no Indigenous with this surname. The surname 高 (Gaō) however, showed up with a significant high-frequency as a Holo surname of Danshui.

Similarly, in 1875, the surname 潘 (Pān) was, according to Ino (1997) assigned to Indigenous groups of Hengchun, Pingdong. The census of 1956 lists only 324 Indigenous of Hengchun with that surname. As a Holo surname, however, 潘 (Pān) appears in the census of Hengchun with a significant high-frequency. A short segment of the census of 1956 for Hengchun is listed below.

Rank	Surname	Fukien (expected)	Indigenous
26	呂 (Lv3)	39 (50)	0
27	何 (Hé)	4 (41)	0
28	蕭 (Xiaō)	9 (47)	0
29	羅 (Luó)	5 (23)	0
30	高 (Gaō)	11 (41)	0
31	潘 (Pān)	69 (32)	81
32	簡 (Jiǎn)	0 (46)	0
33	朱 (Zhū)	112 (26)	0
34	詹 (Zhān)	43 (25)	0
35	游 (Yoú)	0 (36)	0
36	彭 (Péng)	3 (7)	0
37	鍾 (Zhōng)	4 (10)	0
38	施 (Shī)	21 (34)	0
39	沈 (Shěn)	13 (29)	0

This parallel presence of surnames in the Indigenous and Holo population can be explained by an acculturation process, during which Indigenous assimilated Han languages and Han life styles to the point that they and their social environment considered them to be Han. The surname has survived this transformation unaffectedly, even if it was passed down, before full sinification, along maternal lines. An alternative hypothesis which could explain this marked parallel distribution of surnames in two ethnically different populations is that in the case of Hengchun, exceptionally many Holo surnamed 潘 (Pān) moved to Hengchun and most Indigenous surnamed 潘 (Pān) moved away from Hengchun. A question we will tackle here is how probable such hypotheses are. An further alternative is that the Holo in 1956 had still a Pepo identity, but were classified only by local officials as Holo, due to their lifestyle and language use.

To sum up, the census data of 1956 features frequently local configurations that are characterized by the relative over-represented of one surname in two different ethnicities of the same townships. Through this covariation, it should be possible to identify at least some of the Holo surnames of specific townships that go back to acculturated Indigenous groups, without knowing in detail which surnames had been assigned to Indigenous groups more than a century ago.

The approach presented here as Model I resembles to some extent our earlier approach to extract at least some Holo surnames that bear traces of an acculturation process. To achieve this, we first select Holo surnames that in a townships are significantly more frequent than in other townships of the same country.[8] We thus take as a base for the comparison, also in what follows the second level and compare the township to the country. Using as a base for the comparison the entire island, would yield much more extracted names, as much more names would be marked as having an exceptionally high frequency. As we don't have a good criterion for the selection of the one or the other method, we select that one that is to our disadvantage. It is thus impossible to claim that the surnames we extracted are due to the fact that we compared townships with the entire island.

To estimate the probabilities that these unexpectedly high frequent surnames may have been the result of an acculturation process versus a migration, we use resampling methods.[9] We thus model

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9 Most of 20th century descriptive statistics is the result of the need to estimate and approximate values in the absence of sufficient computational power. The speed of modern personal computers, however, allows to let the computer draw samples and just count the percentage of drawings with an outcome that fulfills the expected requirement. Such methods not only allow to get rid of assumptions about distributions, they also allow to easily model complex relations, see Good (1999/2006). In our experiments we repeated each

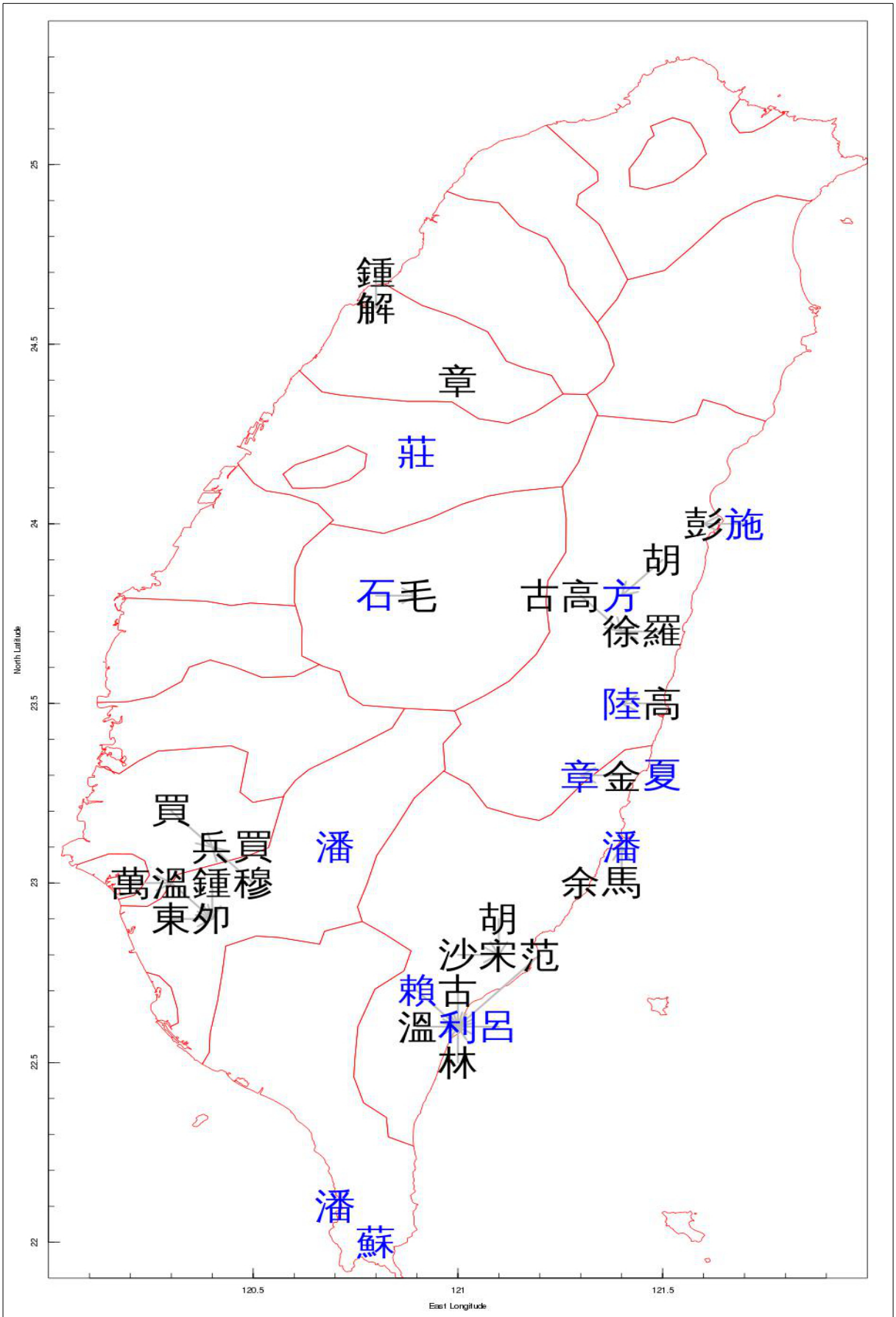
the transition of an older Hakka or Pepo community to a Holo community in 1956, or we model the migration of Holo from townships of the same county into the township. While doing this, we make the simplifying assumption, that the communities of departure had similar statistical properties as the community in 1956. We thus make a copy of the communities of 1956 to have a model of earlier communities.

The estimation of the probability that local peaks in surname distributions are the result of a migration versus an acculturation, we draw randomly persons from the Indigenous communities of departure to construct a new Holo community of arrival of the size of the Holo community of 1956, replacing each drawn person in the community of departure by a person with the same surname. If 97.5% of the sampling experiments result for the surnames in questions in a frequency of a surname not lower than that of the Holo community of 1956, we cannot exclude the hypothesis, that acculturation accounts for the current distribution. In addition, we perform a sampling from the surrounding Holo communities to see whether the Holo community of 1956 can be the result of a migration from the surrounding areas. If less than 2, 5% of the 1000 samplings create a Holo community with the required frequency of a surname, we abandon the hypothesis that migration can have contributed to the observed unexpected frequency of Holo surnames.

Results Model I

The map in Figure locates those Holo and Hakka surnames of the 1956 census that in their local township show a significantly high frequency compared to the respective Holo or Hakka average in Taiwan and the high frequency of which cannot be explained by a migration to that township, but by an acculturation of the local Indigenous population. Two lists of all extracted surnames that pass both statistical tests are given below, first the list of Hakka surnames and second the list of Holo surnames.

sampling 1000 times and counted the number of times the random sampling, modeling a migration or modeling an acculturation, yielded surname frequencies that were equal to or higher than the surname frequencies in *e56*.



4. Figure: Locally unexpectedly high frequent Holo and Hakka surnames of the 1956 census, the local covariance of which with Indigenous surnames is a result of acculturation ($p > .975$) and not the result of migration ($p < .025$. A frequently appearing surname on the map along the west-coast (Qing colonialization) is 潘 (Pān), along the east-coast (Japanese colonialization), the surnames 張 (Zhāng) and 林 (Lín).

Model I: Indigenous-to-Hakka acculturation:

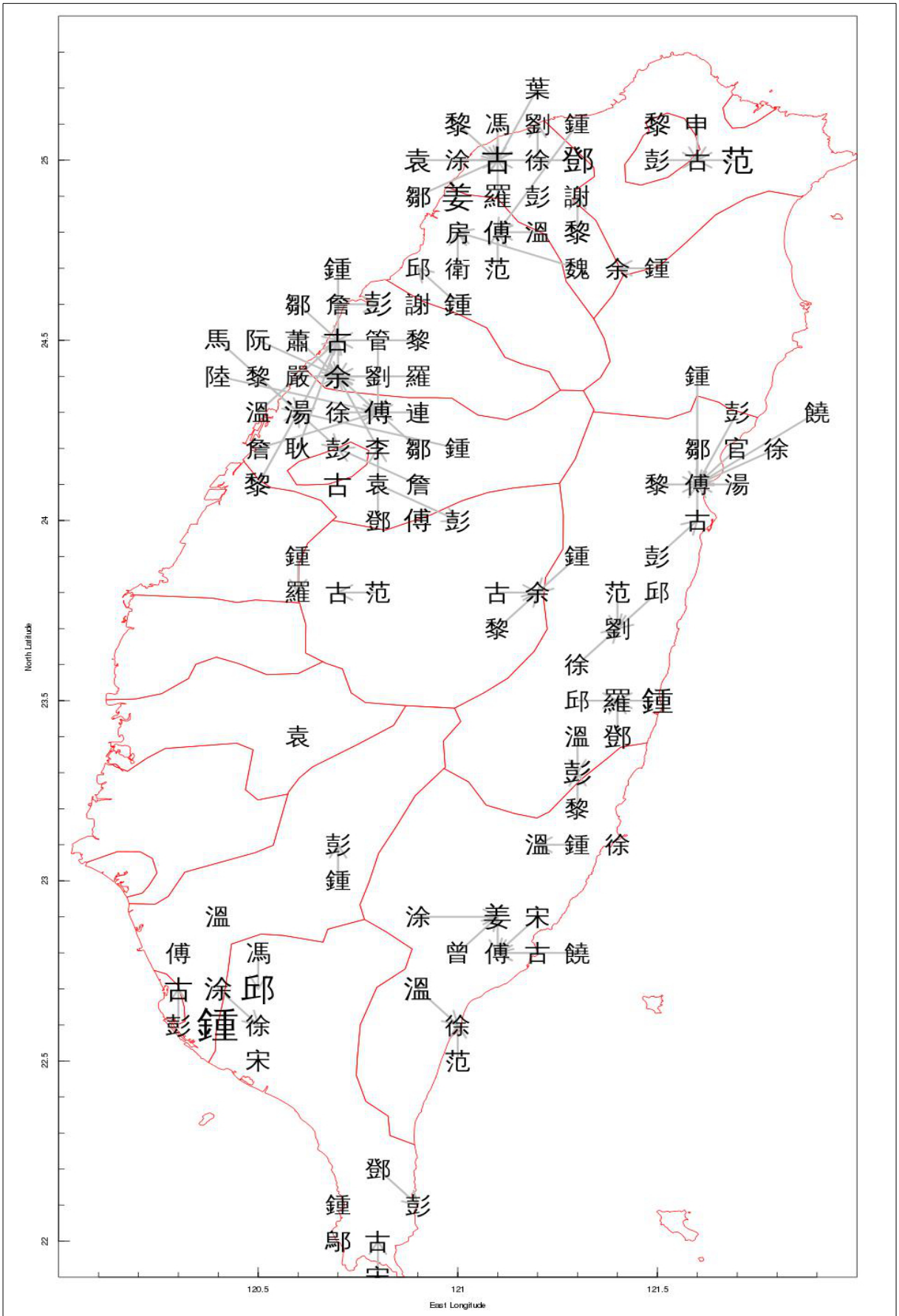
- 台灣, 台灣南部, 屏東市 + 屏東縣, 屏東, 滿州鄉 Pingdongxian: 潘 (Pān), 蘇 (Sū);
- 台灣, 台灣南部, 屏東市 + 屏東縣, 屏東, 車城 Pingdongxian: 潘 (Pān);

- 台灣, 台灣東部, 台東, 太麻里 Taidongxian: 利 (Lì), 呂 (Lv3), 賴 (Laì);
- 台灣, 台灣南部, 后沙, 高雄, 六龜 Gaoxiongxian: 潘 (Pān);
- 台灣, 台灣東部, 台東, 成功 Taidongxian: 潘 (Pān);
- 台灣, 台灣東部, 花蓮, 玉里 Hualianxian: 章 (Zhāng);
- 台灣, 台灣東部, 台東, 長濱 Taidongxian: 夏 (Xià);
- 台灣, 台灣東部, 花蓮, 瑞？ Hualianxian: 陸 (Lù);
- 台灣, 台灣東部, 花蓮, 鳳林 Hualianxian: 方 (Fàng);
- 台灣, 台灣中部, 南投, 魚池 Nantouxian: 石 (Shí);
- 台灣, 台灣東部, 花蓮, 吉安 Hualianxian: 施 (Shī);
- 台灣, 台灣中部, --, 台中, 和平 Taizhongxian: 莊 (Zhuāng);

Model I: Indigenous-to-Holo acculturation:

- 台灣, 台灣東部, 台東, 太麻里 Taidongxian: 古 (Gǔ), 范 (Fàn), 溫 (Wēn), 林 (Lín);
- 台灣, 台灣東部, 台東, 卑南 Taidongxian: 沙 (Shā), 胡 (Hú);
- 台灣, 台灣東部, 台東, 台東 Taidongshi: 宋 (Sòng);
- 台灣, 台灣南部, 后沙, 高雄, 內門 Gaoxiongxian: 邶 (Maǒ), 東 (Dōng), 溫 (Wēn), 鍾 (Zhōng);
- 台灣, 台灣東部, 台東, 東河 Taidongxian: 余 (Yú);
- 台灣, 台灣南部, 台南市 + 台南縣, 台南, 新化 Tainanxian: 萬 (Wàn);
- 台灣, 台灣東部, 台東, 成功 Taidongxian: 馬 (Mǎ);
- 台灣, 台灣南部, 台南市 + 台南縣, 台南, 玉井 Tainanxian: 穆 (Mù), 兵 (Bīng), 買 (Maǐ);
- 台灣, 台灣南部, 台南市 + 台南縣, 台南, 楠西 Tainanxian: 買 (Maǐ);
- 台灣, 台灣東部, 花蓮, 玉里 Hualianxian: 金 (Jīn);
- 台灣, 台灣東部, 花蓮, 瑞？ Hualianxian: 高 (Gaō);
- 台灣, 台灣東部, 花蓮, 光復鄉 Hualianxian: 高 (Gaō), 徐 (Xú), 羅 (Luó);
- 台灣, 台灣東部, 花蓮, 鳳林 Hualianxian: 胡 (Hú);
- 台灣, 台灣中部, 南投, 魚池 Nantouxian: 毛 (Maó);
- 台灣, 台灣東部, 富里鄉: 古 (Gǔ);
- 台灣, 台灣東部, 花蓮, 吉安 Hualianxian: 彭 (Péng);
- 台灣, 台灣東部, 苗栗, 苗栗, 南庄 Miaolixian: 章 (Zhāng);
- 台灣, 台灣東部, 苗栗, 苗栗, 後龍 Miaolixian: 解 (Xiè), 鍾 (Zhōng);

The map in Figure locates those Holo surnames of the 1956 census that in their local township show a significantly high frequency compared to the Holo average and the high frequency of which cannot be explained by a migration to that locality, but by an acculturation of the local Hakka population. A list of surnames that pass both statistical criteria is given below.



5. Figure: Locally unexpectedly high frequent Holo surnames of the 1956 census, the local covariance of which with Hakka surnames is a result of acculturation ($p > .975$) and not the result of migration ($p < .025$).

Model I: Hakka-to-Holo acculturation:

- 台灣, 台灣南部, 屏東市 + 屏東縣, 屏東, 恆春 Pingdongxian: 鄔 (Wū);
- 台灣, 台灣南部, 屏東市 + 屏東縣, 屏東, 滿州鄉 Pingdongxian: 宋 (Sòng), 古 (Gǔ);
- 台灣, 台灣南部, 屏東市 + 屏東縣, 屏東, 車城 Pingdongxian: 鍾 (Zhōng);
- 台灣, 台灣東部, 宜蘭, 大同 Yilanxian: 彭 (Péng), 鄧 (Dèng);

- 台灣, 台灣南部, 屏東市 + 屏東縣, 屏東, 佳冬 Pingdongxian: 鍾 (Zhōng);
- 台灣, 台灣南部, 屏東市 + 屏東縣, 屏東, 南洲 Pingdongxian: 宋 (Sòng);
- 台灣, 台灣南部, 屏東市 + 屏東縣, 屏東, 潮州 Pingdongxian: 徐 (Xú), 鍾 (Zhōng), 邱 (Qiū), 涂 (Tú);
- 台灣, 台灣南部, 后沙, 高雄, 小港 Gaoxiongshi: 鍾 (Zhōng);
- 台灣, 台灣東部, 台東, 太麻里 Taidongxian: 范 (Fàn), 徐 (Xú), 溫 (Wēn);
- 台灣, 台灣南部, 后沙, 高雄, 三民 Gaoxiongshi: 彭 (Péng);
- 台灣, 台灣南部, 后沙, 高雄, 苓雅區 Gaoxiongxian: 古 (Gǔ);
- 台灣, 台灣南部, 后沙, 高雄, 寶珠 Gaoxiongshi: 古 (Gǔ);
- 台灣, 台灣南部, 屏東市 + 屏東縣, 屏東 Pindongshi: 邱 (Qiū), 鍾 (Zhōng);
- 台灣, 台灣南部, 屏東市 + 屏東縣, 屏東, 九如 Pingdongxian: 馮 (Féng), 鍾 (Zhōng);
- 台灣, 台灣南部, 屏東市 + 屏東縣, 屏東, 鹽埔 Pingdongxian: 邱 (Qiū);
- 台灣, 台灣南部, 后沙, 高雄, 橋頭 Gaoxiongxian: 傅 (Fù);
- 台灣, 台灣東部, 台東, 卑南 Taidongxian: 饒 (Raó), 姜 (Jiāng);
- 台灣, 台灣南部, 屏東市 + 屏東縣, 屏東, 里港 Pingdongxian: 鍾 (Zhōng);
- 台灣, 台灣東部, 台東, 台東 Taidongshi: 古 (Gǔ), 溫 (Wēn), 宋 (Sòng), 姜 (Jiāng), 傅 (Fù);
- 台灣, 台灣南部, 屏東市 + 屏東縣, 屏東, 高樹 Pingdongxian: 邱 (Qiū), 涂 (Tú);
- 台灣, 台灣南部, 后沙, 高雄, 杉林 Gaoxiongxian: 溫 (Wēn);
- 台灣, 台灣東部, 台東, 鹿野 Taidongxian: 曾 (Zēng), 涂 (Tú);
- 台灣, 台灣南部, 后沙, 高雄, 六龜 Gaoxiongxian: 鍾 (Zhōng), 彭 (Péng);
- 台灣, 台灣東部, 台東, 成功 Taidongxian: 徐 (Xú);
- 台灣, 台灣東部, 台東, 池上 Taidongxian: 鍾 (Zhōng), 溫 (Wēn);
- 台灣, 台灣東部, 花蓮, 玉里 Hualianxian: 黎 (Lí), 溫 (Wēn), 彭 (Péng);
- 台灣, 台灣東部, 台東, 長濱 Taidongxian: 彭 (Péng);
- 台灣, 台灣南部, 嘉義市 + 嘉義縣, 嘉義, 中埔鄉 Jiayixian: 袁 (Yuán);
- 台灣, 台灣東部, 花蓮, 瑞？ Hualianxian: 鄧 (Dèng), 鍾 (Zhōng), 邱 (Qiū), 羅 (Luó);
- 台灣, 台灣東部, 花蓮, 光復鄉 Hualianxian: 鍾 (Zhōng), 羅 (Luó), 邱 (Qiū), 范 (Fàn), 徐 (Xú);
- 台灣, 台灣東部, 花蓮, 壽豐鄉 Yilanxian: 劉 (Liú), 鄧 (Dèng), 鍾 (Zhōng);
- 台灣, 台灣中部, 南投, 竹山 Nantouxian: 范 (Fàn), 古 (Gǔ);
- 台灣, 台灣南部, 后沙, 高雄, 甲仙 Gaoxiongxian: 羅 (Luó), 鍾 (Zhōng);
- 台灣, 台灣中部, 南投, 魚池 Nantouxian: 傅 (Fù);
- 台灣, 台灣東部, 富里鄉: 余 (Yú), 鍾 (Zhōng), 黎 (Lí), 古 (Gǔ);
- 台灣, 台灣中部, 南投, 埔里 Nantouxian: 彭 (Péng), 袁 (Yuán);
- 台灣, 台灣東部, 花蓮, 吉安 Hualianxian: 彭 (Péng);
- 台灣, 台灣中部, 南投, 水裡 Nantouxian: 傅 (Fù);
- 台灣, 台灣東部, 花蓮, 花蓮 Hualianxian: 湯 (Tāng), 古 (Gǔ), 鄒 (Zoú), 徐 (Xú), 官 (Guān), 黎 (Lí), 饒 (Raó), 鍾 (Zhōng), 彭 (Péng);
- 台灣, 台灣東部, 花蓮, 新城 Hualianxian: 傅 (Fù);
- 台灣, 台灣中部, --, 台中, 霧峯 Taizhongxian: 古 (Gǔ), 黎 (Lí);
- 台灣, 台灣中部, --, 台中, 后里 Taizhongxian: 詹 (Zhān), 彭 (Péng), 黎 (Lí);
- 台灣, 台灣中部, --, 台中, 新社 Taizhongxian: 鄧 (Dèng);
- 台灣, 台灣中部, --, 台中, 豐原 Taizhongxian: 徐 (Xú), 傅 (Fù), 連 (Lián), 古 (Gǔ), 陸 (Lù), 管 (Guǎn), 詹 (Zhān);
- 台灣, 台灣中部, --, 台中, 外埔 Taizhongxian: 馬 (Mǎ), 鍾 (Zhōng);
- 台灣, 台灣中部, --, 台中, 大甲 Taizhongxian: 黎 (Lí);
- 台灣, 台灣東部, 苗栗, 苗栗, 苑裡 Miaolixian: 鄒 (Zoú), 鍾 (Zhōng), 羅 (Luó), 李 (Lǐ), 余 (Yú), 古 (Gǔ), 湯 (Tāng), 嚴 (Yán), 劉 (Liú), 蕭 (Xiāo), 阮 (Ruǎn);
- 台灣, 台灣東部, 苗栗, 苗栗, 通霄 Miaolixian: 古 (Gǔ), 湯 (Tāng), 鄒 (Zoú), 溫 (Wēn), 耿 (Gěng), 黎 (Lí);
- 台灣, 台灣東部, 苗栗, 苗栗 Miaolishi: 傅 (Fù);
- 台灣, 台灣東部, 苗栗, 苗栗, 造橋 Miaolixian: 謝 (Xiè);
- 台灣, 台灣東部, 苗栗, 苗栗, 後龍 Miaolixian: 余 (Yú), 鍾 (Zhōng), 彭 (Péng), 詹 (Zhān);
- 台灣, 台灣北部, 逃園, 復興鄉 Taoyuanxian: 鍾 (Zhōng), 鄧 (Dèng), 余 (Yú);
- 台灣, 台灣東部, 苗栗, 苗栗, 竹南 Miaolixian: 鍾 (Zhōng), 邱 (Qiū);
- 台灣, 台灣北部, 新竹, 新竹, 竹北 Xinzhuxian: 衛 (Wèi), 姜 (Jiāng), 鍾 (Zhōng), 房 (Fáng),

彭 (Péng), 魏 (Weì);
- 台灣, 台灣北部, 新竹, 新竹 Xinzhushi: 鍾 (Zhōng), 傅 (Fù), 范 (Fàn), 溫 (Wēn), 姜 (Jiāng);
- 台灣, 台灣北部, 逃園, 龍潭 Taoyuanxian: 彭 (Péng);
- 台灣, 台灣北部, 逃園, 大溪 Taoyuanxian: 古 (Gǔ), 傅 (Fù), 黎 (Lí);
- 台灣, 台灣北部, 新竹, 新竹, 新豐 Xinzhuxian: 姜 (Jiāng);
- 台灣, 台灣北部, 逃園, 八德 Taoyuanxian: 謝 (Xiè);
- 台灣, 台灣北部, 逃園, 桃園 Taoyuanxian: 黎 (Lí), 古 (Gǔ);
- 台灣, 台灣北部, 逃園, 中壢 Taoyuanxian: 鄧 (Dèng), 涂 (Tú), 黎 (Lí), 馮 (Féng), 古 (Gǔ), 袁 (Yuán), 羅 (Luó), 鄒 (Zoú);
- 台灣, 台灣北部, 逃園, 中壢, 觀音鄉 Taoyuanxian: 葉 (Yè), 羅 (Luó);
- 台灣, 台灣北部, 台北, 台北, 龍山 Taibeishi: 申 (Shēn);
- 台灣, 台灣北部, 台北, 台北, 延平 Taibeishi: 范 (Fàn);
- 台灣, 台灣北部, 台北, 台北, 城中 Taibeishi: 范 (Fàn);
- 台灣, 台灣北部, 台北, 台北, 建成 Taibeishi: 范 (Fàn);
- 台灣, 台灣北部, 台北, 台北, 松山 Taibeishi: 彭 (Péng), 古 (Gǔ);
- 台灣, 台灣北部, 逃園, 大園 Taoyuanxian: 黎 (Lí), 徐 (Xú), 劉 (Liú), 袁 (Yuán);
- 台灣, 台灣北部, 台北, 台北, 三重 Taibeixian: 黎 (Lí);
- 台灣, 台灣北部, 台北, 台北, 大同 Taibeishi: 黎 (Lí), 鄧 (Dèng);

Evaluation adn Discussion of Model 1

In order to obtain an approximate impression of what the surnames extracted by Model I identify, we make use of the ThakBong tombstone archive. If the surnames extracted are be special in the sense that they make it likely that the ancestors of the bearer of these surnames changed their ethnicity, e.g. from Indigenous to Holo, then we might expect that the tombs in our sample are also special, either by their features of by their local distribution. As despite the size of about 30.000 tombs in the ThakBong archive, the likelihood to find tombs of low-frequency surnames in specific regions is relatively low, we have to restrict the attempt to evaluate the outcomes of Model I to the regions where we sampled most extensively, that is Tainan and Kaohsiung.

A surname extracted as indicator of an Indigenous to Holo acculturation is 買 (Maǐ). In the ThakBong archive, six tombs with this surname are located in Tainan, two in Gaoxiong. Of these, seven are Christian tombs, the 8th tomb is in Shanshang, close to Yujing, Tainan, a place where Siraya reside. As a baseline of comparison, in Tainan 12% of a sample of about 7000 Holo tombs are Christian. The high percentage of Christians in this small sample confirms the hypothesis of an Indigenous origin, as Indigenous communities have been much more receptive to Christian doctrines and the mass conversion took place in the second half of the 19th century. Still today, churches, and especially old churches in Taiwan's rural areas can be considered to be indicators of Indigenous roots of a community. Among the examples below, only the first tombstone below would count as a typical Holo tombstone, given the shape of the tombstones, the number of characters, the symbols and the absence of photos.

Of the three Holo tombs in Tainan of persons surnamed 卓 (Zhuó), two tombs are Christian. The given name of the deceased of the third, non-Christian tomb contains the character 番 (fān) [Barbarian], associated with the ”Wilderness” (荒服 huāng1fú), the outmost zone in the Chinese cosmographic conception of the world as defined in the “Tribute of Yu” (禹貢, Yǔ3gòng) which is inhabited by people which are most different from Chinese people.

Six of the seven tombs of persons surnamed 穆 (Mù) in our tombstone sample from Taiwan are located in Tainan. Six of the seven are Christian tombs, the seventh tomb is in Xinshi, few kilometers from Xinhua.

Six of the thirteen Holo tombs in Tainan of persons surnamed 戴 (Daì) are Christian.

Model I, proposed to gauge the acculturation processes on the basis of the census of 1956. This model first of all fulfills the requirement to produced probabilities which can be used for further modeling. Second, the model passed a face-validation, using as criterion the distribution and features of tombs in the ThakBong Archive that are associated with the extracted surnames. This however can only the beginning of a research. First, for different periods and different socio-cultural environment, different models would be required. The proposed model, for example, requires the presence in 1956 of Indigenous groups within a township in order to be applicable. If no Indigenous

with a certain surname are left, if for example the acculturation process is already completed, this Model I has to be replaced or completed with other approaches. The approach presented in Streiter, Goudin, Huang and Lin (2011), imperfect as it might be, does not require such a remaining population and would thus potentially also apply for earlier phases of surnaming and acculturation. Second, and equally important, this model has still a number of flaws which successively should and can be remedied in the future.

First, this modeling of the transition of local communities from a certain point in the past to the local communities of 1956 is not anchored to some concrete time in the past. From the similarity of the few known cases of surnaming, such as the surnaming of Indigenous in Hengchun as described in Ino (1997), we can only guess, that the process modeled by Model I spans the time from late Qing, e.g. 1870 to 1956. However, no formal criteria link the data to that time period. Second, Model I uses as a template for the communities of times before 1956 a perfect copy of the data of 1956. This, of course, is questionable for various reasons. Local peaks of surnames are leveled out through time through migraton, as shown for the case of Nanhua by Shi (1995). Of the modern Indigenous Pe-po population of Nanhua, two third have been migrated to that region between the Japanese colonizations and the 1990s. A community that had originally 13 李 (Lǐ) and 1 買 (Maǐ), had in the 1990s also 1 陳 (Chén), 1 王 (Wáng), 1 毛 (Maó), 5 卓 (Zhuó)and 1 溫 (Wēn). If migration tends to level out local surname particularities, we can assume, that if in 1956 community had a marked surname distribution, 100 years before, this distribution must have been even more marked. Fiurther differences relate to the different proportions of ethnicities in communities in different times and different frequency relations compared to other communities.

Fifth, acculturation and migration have been treated in Model I as alternative or even concurring processes. In reality, however, they are simultaneously ongoing and conditioning each other mutually. The census conducted by the Japanese authorities in 1928 lists no Han population in the administrative units listed below. If no Han resided in those regions, there was also no or little pressure on the Indigenous population to assimilate to Han life styles. The demographic processes between 1928 and 1956 in these regions was thus necessarily a combination of migration of Han to those regions and, depending on it, an acculturation. Migration on the other hand depends on a process of acculturation. If in 1928 virtually no Han had migrated into these regions, it was also because the majoritarian Indigenous had no understanding for Han life-styles and would not welcome new settlers.

Japanese	Today
大武支廳 (臺東廳)	台灣, 台灣東部, 台東
新港支廳 (臺東廳)	台灣, 台灣東部, 台東
卑南區 (臺東支廳)	台灣, 台灣東部, 台東, 卑南
大武區 (大武支廳)	台灣, 台灣東部, 台東, 大武
太麻里區 (大武支廳)	台灣, 台灣東部, 台東, 太麻里
新港區 (新港支廳)	台灣, 台灣東部, 台東, 成功
成廣澳區 (新港支廳)	台灣, 台灣東部, 台東, 成功
都鑾區 (新港支廳)	台灣, 台灣東部, 台東, 成功
加走灣區 (新港支廳)	台灣, 台灣東部, 台東, 成功
鹿野區 (里壠支廳)	台灣, 台灣東部, 台東, 鹿野
研海支廳 (花蓮港廳)	台灣, 台灣東部, 花蓮
研海區 (研海支廳)	台灣, 台灣東部, 花蓮, 新城
新社區 (花蓮支廳)	台灣, 台灣東部, 花蓮, 豐濱

Model II

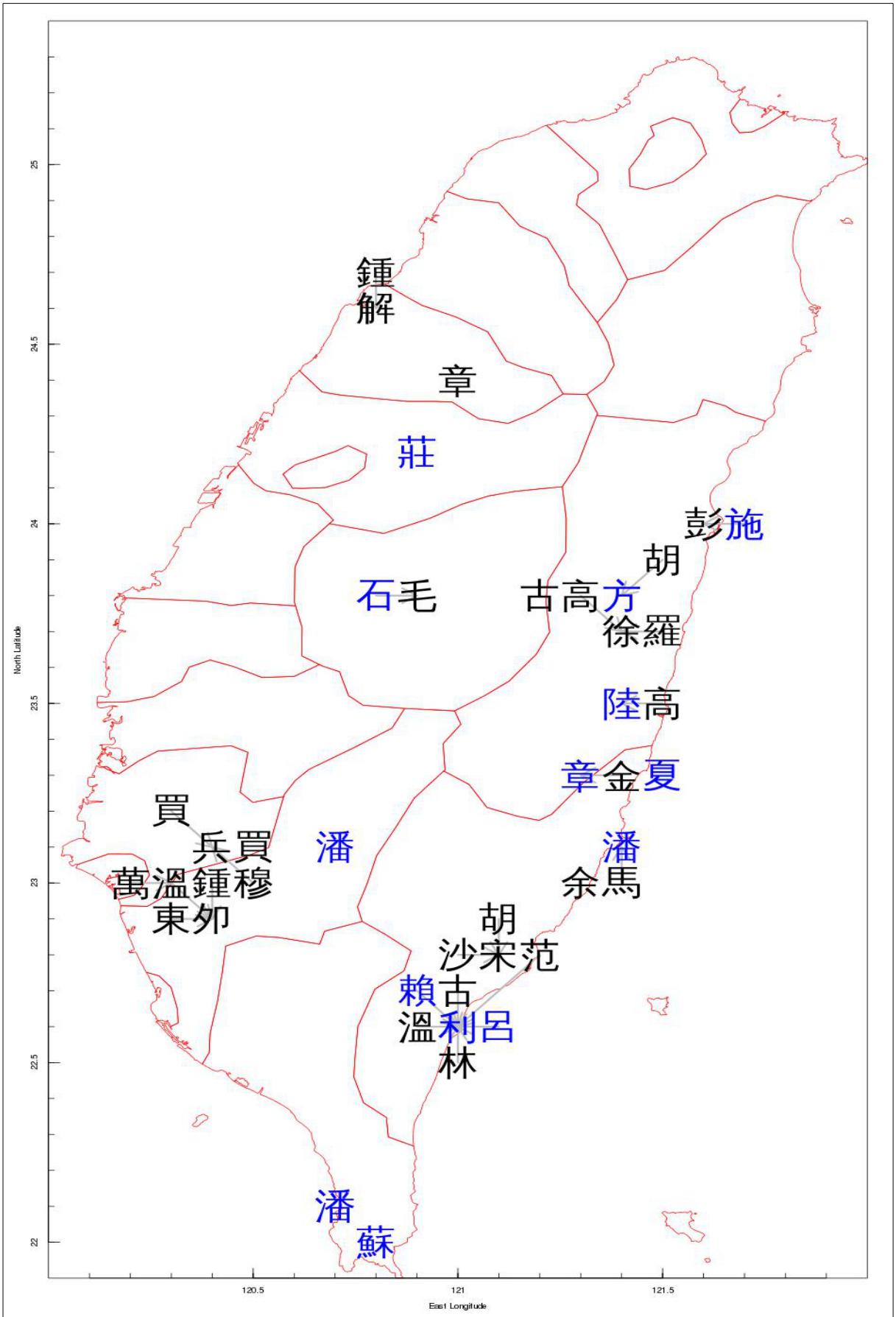
To show how the approach described and tested as Model I can be successively improved, we apply here one correction, concerning the relative population size of ethnicities at 1956 and 1928, as a random point in time of which we have solid population statistics. We estimate the size of the Indigenous population at that time to be about 600.000.[10] The size of the Holo and Hakka population can be assumed to equal, according to the Japanese census of 1928, 3.000.000 and 600.000 respectively. The census of 1956 lists for these ethnic groups 220.000, 7.0000.000 and 1.220.000 respectively. To establish an estimation of the population of 1928 on the basis of the data of 1956, we apply factors to the distribution of 1956: 2.72 for the Indigenous groups, 0.43 for the Holo and

10 The Japanese census of 1928 gives the number of 3751600 for the Holo population and the data of Xu, Chen, Huang and Huang (2010) sum up to 4346090 for the entire population of that years. The number of 600.000 seems reasonable, as the Japanese census of 1905 specifies the number of the Indigenous as above 750.000.

0.49 for the Hakka. Estimated, there are thus about 180.000 Hakka classified in 1928 as Hakka who in 1956 were classified as Holo. Similarly there is an estimated number of 900.000 people who in 1928 where classified as Indigenoius and 1956 were classified as Holo. Through this augmentation of the relative size of the Indigenous and Hakka population with respect to the Holo population, probabilities estimated for the assimilation process might thus become more likely.

Results Model I

The map in Figure locates those Holo and Hakka surnames of the 1956 census that in their local township show a significant high-frequency compared to the respective Holo or Hakka average in Taiwan and the high frequency of which cannot be explained by a migration to that township, but by an acculturation of the local Indigenous population. Two lists of all extracted surnames that pass both statistical tests are given below, first the list of Hakka surnames and second the list of Holo surnames. Compared to Model I ...



6. Figure: Model II: Locally unexpectedly high frequent Holo and Hakka surnames of the 1956 census, the local covariance of which with Indigenous surnames is a result of acculturation ($p > .975$) and not the result of migration ($p < .025$.

Model II: Indigenous-to-Hakka acculturation:

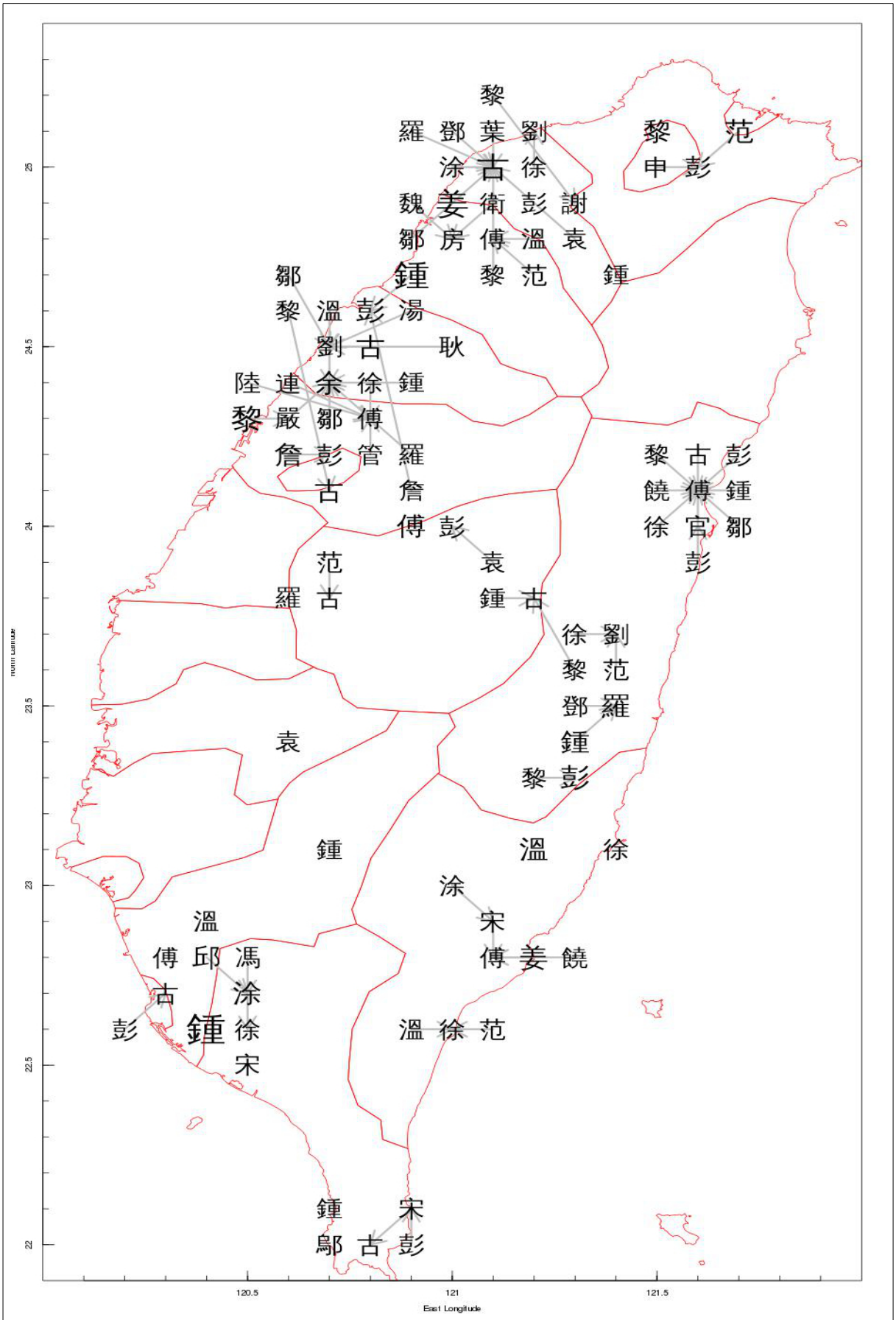
- 台灣, 台灣南部, 屏東市 + 屏東縣, 屏東, 滿州鄉 Pingdongxian: 潘 (Pān), 蘇 (Sū);
- 台灣, 台灣南部, 屏東市 + 屏東縣, 屏東, 車城 Pingdongxian: 潘 (Pān);
- 台灣, 台灣東部, 宜蘭, 大同 Yilanxian: 葉 (Yè);
- 台灣, 台灣南部, 屏東市 + 屏東縣, 屏東, 萬巒: 潘 (Pān);

- 台灣, 台灣東部, 台東, 太麻里 Taidongxian: 利 (Lì), 呂 (Lv3), 賴 (Laì);
- 台灣, 台灣南部, 后沙, 高雄, 六龜 Gaoxiongxian: 潘 (Pān);
- 台灣, 台灣東部, 台東, 成功 Taidongxian: 潘 (Pān), 林 (Lín), 宋 (Sòng);
- 台灣, 台灣東部, 花蓮, 玉里 Hualianxian: 陸 (Lù), 章 (Zhāng);
- 台灣, 台灣東部, 台東, 長濱 Taidongxian: 邱 (Qiū), 夏 (Xià);
- 台灣, 台灣南部, 嘉義市 + 嘉義縣, 嘉義, 吳鳳 Jiayixian: 杜 (Dù);
- 台灣, 台灣東部, 花蓮, 瑞？ Hualianxian: 陸 (Lù), 馬 (Mǎ);
- 台灣, 台灣東部, 花蓮, 鳳林 Hualianxian: 方 (Fàng);
- 台灣, 台灣南部, 后沙, 高雄, 甲仙 Gaoxiongxian: 呂 (Lv3);
- 台灣, 台灣中部, 南投, 魚池 Nantouxian: 石 (Shí);
- 台灣, 台灣東部, 花蓮, 吉安 Hualianxian: 高 (Gaō), 趙 (Zhaò), 施 (Shī);
- 台灣, 台灣中部, 南投, 仁愛 Nantouxian: 鄭 (Zhèng);
- 台灣, 台灣中部, --, 台中, 和平 Taizhongxian: 莊 (Zhuāng);
- 台灣, 台灣東部, 宜蘭, 南澳 Yilanxian: 許 (Xǔ);

Model II: Indigenous-to-Holo acculturation:

- 台灣, 台灣東部, 台東, 太麻里 Taidongxian: 古 (Gǔ), 范 (Fàn), 溫 (Wēn), 林 (Lín);
- 台灣, 台灣東部, 台東, 卑南 Taidongxian: 沙 (Shā), 胡 (Hú);
- 台灣, 台灣東部, 台東, 台東 Taidongshi: 宋 (Sòng);
- 台灣, 台灣南部, 后沙, 高雄, 內門 Gaoxiongxian: 夘 (Maǒ), 東 (Dōng), 鍾 (Zhōng);
- 台灣, 台灣南部, 台南市 + 台南縣, 台南, 新化 Tainanxian: 萬 (Wàn);
- 台灣, 台灣東部, 台東, 成功 Taidongxian: 馬 (Mǎ);
- 台灣, 台灣南部, 台南市 + 台南縣, 台南, 玉井 Tainanxian: 穆 (Mù), 兵 (Bīng), 買 (Maǐ);
- 台灣, 台灣南部, 台南市 + 台南縣, 台南, 楠西 Tainanxian: 買 (Maǐ);
- 台灣, 台灣東部, 花蓮, 玉里 Hualianxian: 金 (Jīn);
- 台灣, 台灣東部, 花蓮, 光復鄉 Hualianxian: 徐 (Xú), 羅 (Luó);
- 台灣, 台灣東部, 花蓮, 鳳林 Hualianxian: 胡 (Hú);
- 台灣, 台灣東部, 富里鄉: 古 (Gǔ);
- 台灣, 台灣東部, 花蓮, 吉安 Hualianxian: 彭 (Péng);
- 台灣, 台灣東部, 苗栗, 苗栗, 南庄 Miaolixian: 章 (Zhāng);
- 台灣, 台灣東部, 苗栗, 苗栗, 後龍 Miaolixian: 解 (Xiè), 鍾 (Zhōng);

The map in Figure locates those Holo surnames of the 1956 census that in their local township show a significantly high frequency compared to the Holo average and the high frequency of which cannot be explained by a migration to that locality, but by an acculturation of the local Hakka population. A list of surnames that pass both statistical criteria is given below. Compared to Model I ...



7. Figure: Model II: Locally unexpectedly high frequent Holo surnames of the 1956 census, the local covariance of which with Hakka surnames is a result of acculturation ($p > .975$) and not the result of migration ($p < .025$).

Model II: Hakka-to-Holo acculturation:

- 台灣, 台灣南部, 屏東市 + 屏東縣, 屏東, 恆春 Pingdongxian: 鄔 (Wū);
- 台灣, 台灣南部, 屏東市 + 屏東縣, 屏東, 滿州鄉 Pingdongxian: 宋 (Sòng), 古 (Gǔ);
- 台灣, 台灣南部, 屏東市 + 屏東縣, 屏東, 車城 Pingdongxian: 鍾 (Zhōng);
- 台灣, 台灣東部, 宜蘭, 大同 Yilanxian: 彭 (Péng);

- 台灣, 台灣南部, 屏東市 + 屏東縣, 屏東, 佳冬 Pingdongxian: 鍾 (Zhōng);
- 台灣, 台灣南部, 屏東市 + 屏東縣, 屏東, 南洲 Pingdongxian: 宋 (Sòng);
- 台灣, 台灣南部, 屏東市 + 屏東縣, 屏東, 潮州 Pingdongxian: 徐 (Xú), 鍾 (Zhōng), 涂 (Tú);
- 台灣, 台灣南部, 后沙, 高雄, 小港 Gaoxiongshi: 鍾 (Zhōng);
- 台灣, 台灣東部, 台東, 太麻里 Taidongxian: 范 (Fàn), 徐 (Xú), 溫 (Wēn);
- 台灣, 台灣南部, 后沙, 高雄, 三民 Gaoxiongshi: 彭 (Péng);
- 台灣, 台灣南部, 后沙, 高雄, 寶珠 Gaoxiongshi: 古 (Gǔ);
- 台灣, 台灣南部, 屏東市 + 屏東縣, 屏東 Pindongshi: 邱 (Qiū);
- 台灣, 台灣南部, 屏東市 + 屏東縣, 屏東, 九如 Pingdongxian: 馮 (Féng), 鍾 (Zhōng);
- 台灣, 台灣南部, 后沙, 高雄, 橋頭 Gaoxiongxian: 傅 (Fù);
- 台灣, 台灣東部, 台東, 卑南 Taidongxian: 饒 (Raó), 姜 (Jiāng);
- 台灣, 台灣南部, 屏東市 + 屏東縣, 屏東, 里港 Pingdongxian: 鍾 (Zhōng);
- 台灣, 台灣東部, 台東, 台東 Taidongshi: 宋 (Sòng), 姜 (Jiāng), 傅 (Fù);
- 台灣, 台灣南部, 屏東市 + 屏東縣, 屏東, 高樹 Pingdongxian: 邱 (Qiū), 涂 (Tú);
- 台灣, 台灣南部, 后沙, 高雄, 杉林 Gaoxiongxian: 溫 (Wēn);
- 台灣, 台灣東部, 台東, 鹿野 Taidongxian: 涂 (Tú);
- 台灣, 台灣南部, 后沙, 高雄, 六龜 Gaoxiongxian: 鍾 (Zhōng);
- 台灣, 台灣東部, 台東, 成功 Taidongxian: 徐 (Xú);
- 台灣, 台灣東部, 台東, 池上 Taidongxian: 溫 (Wēn);
- 台灣, 台灣東部, 花蓮, 玉里 Hualianxian: 黎 (Lí), 溫 (Wēn), 彭 (Péng);
- 台灣, 台灣東部, 台東, 長濱 Taidongxian: 彭 (Péng);
- 台灣, 台灣南部, 嘉義市 + 嘉義縣, 嘉義, 中埔鄉 Jiayixian: 袁 (Yuán);
- 台灣, 台灣東部, 花蓮, 瑞？ Hualianxian: 鄧 (Dèng), 鍾 (Zhōng), 羅 (Luó);
- 台灣, 台灣東部, 花蓮, 光復鄉 Hualianxian: 鍾 (Zhōng), 羅 (Luó), 范 (Fàn), 徐 (Xú);
- 台灣, 台灣東部, 花蓮, 壽豐鄉 Yilanxian: 劉 (Liú);
- 台灣, 台灣中部, 南投, 竹山 Nantouxian: 范 (Fàn), 古 (Gǔ);
- 台灣, 台灣南部, 后沙, 高雄, 甲仙 Gaoxiongxian: 羅 (Luó);
- 台灣, 台灣中部, 南投, 魚池 Nantouxian: 傅 (Fù);
- 台灣, 台灣東部, 富里鄉: 鍾 (Zhōng), 黎 (Lí), 古 (Gǔ);
- 台灣, 台灣中部, 南投, 埔里 Nantouxian: 彭 (Péng), 袁 (Yuán);
- 台灣, 台灣東部, 花蓮, 吉安 Hualianxian: 彭 (Péng);
- 台灣, 台灣中部, 南投, 水裡 Nantouxian: 傅 (Fù);
- 台灣, 台灣東部, 花蓮, 花蓮 Hualianxian: 古 (Gǔ), 鄒 (Zoú), 徐 (Xú), 官 (Guān), 黎 (Lí), 饒 (Raó), 鍾 (Zhōng), 彭 (Péng);
- 台灣, 台灣東部, 花蓮, 新城 Hualianxian: 傅 (Fù);
- 台灣, 台灣中部, --, 台中, 霧峯 Taizhongxian: 古 (Gǔ), 黎 (Lí);
- 台灣, 台灣中部, --, 台中, 后里 Taizhongxian: 詹 (Zhān), 彭 (Péng), 黎 (Lí);
- 台灣, 台灣中部, --, 台中, 豐原 Taizhongxian: 徐 (Xú), 傅 (Fù), 連 (Lián), 古 (Gǔ), 陸 (Lù), 管 (Guǎn), 詹 (Zhān);
- 台灣, 台灣中部, --, 台中, 大甲 Taizhongxian: 黎 (Lí);
- 台灣, 台灣東部, 苗栗, 苗栗, 苑裡 Miaolixian: 鄒 (Zoú), 鍾 (Zhōng), 羅 (Luó), 余 (Yú), 古 (Gǔ), 嚴 (Yán), 劉 (Liú);
- 台灣, 台灣東部, 苗栗, 苗栗, 通霄 Miaolixian: 古 (Gǔ), 湯 (Tāng), 鄒 (Zoú), 溫 (Wēn), 耿 (Gěng), 黎 (Lí);
- 台灣, 台灣東部, 苗栗, 苗栗, 後龍 Miaolixian: 余 (Yú), 鍾 (Zhōng), 彭 (Péng), 詹 (Zhān);
- 台灣, 台灣北部, 逃園, 復興鄉 Taoyuanxian: 鍾 (Zhōng);
- 台灣, 台灣東部, 苗栗, 苗栗, 竹南 Miaolixian: 鍾 (Zhōng);
- 台灣, 台灣北部, 新竹, 新竹, 竹北 Xinzhuxian: 衛 (Weì), 姜 (Jiāng), 鍾 (Zhōng), 房 (Fáng), 彭 (Péng), 魏 (Weì);
- 台灣, 台灣北部, 新竹, 新竹 Xinzhushi: 鍾 (Zhōng), 傅 (Fù), 范 (Fàn), 溫 (Wēn), 姜 (Jiāng);
- 台灣, 台灣北部, 逃園, 龍潭 Taoyuanxian: 彭 (Péng);
- 台灣, 台灣北部, 逃園, 大溪 Taoyuanxian: 古 (Gǔ), 黎 (Lí);
- 台灣, 台灣北部, 新竹, 新竹, 新豐 Xinzhuxian: 姜 (Jiāng);
- 台灣, 台灣北部, 逃園, 八德 Taoyuanxian: 謝 (Xiè);
- 台灣, 台灣北部, 逃園, 桃園 Taoyuanxian: 古 (Gǔ);
- 台灣, 台灣北部, 逃園, 中壢 Taoyuanxian: 鄧 (Dèng), 涂 (Tú), 黎 (Lí), 古 (Gǔ), 袁 (Yuán),

鄒 (Zoú);
- 台灣, 台灣北部, 逃園, 中壢, 觀音鄉 Taoyuanxian: 葉 (Yè), 羅 (Luó);
- 台灣, 台灣北部, 台北, 台北, 龍山 Taibeishi: 申 (Shēn);
- 台灣, 台灣北部, 台北, 台北, 延平 Taibeishi: 范 (Fàn);
- 台灣, 台灣北部, 台北, 台北, 城中 Taibeishi: 范 (Fàn);
- 台灣, 台灣北部, 台北, 台北, 松山 Taibeishi: 彭 (Péng);
- 台灣, 台灣北部, 逃園, 大園 Taoyuanxian: 徐 (Xú), 劉 (Liú);
- 台灣, 台灣北部, 台北, 台北, 三重 Taibeixian: 黎 (Lí);
- 台灣, 台灣北部, 台北, 台北, 大同 Taibeishi: 黎 (Lí);

Summary Discussion

Starting from the need to estimate the likelihood of the ethnicity associated with a deceased and his or her tomb, we investigated the potentials of the census of 1956 for this task.

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